Founder Memo: COD Risk Decision Engine

1. Executive Summary

The objective of this decision engine is to reduce Return to Origin (RTO) losses without unnecessarily rejecting genuine COD customers.

The historical dataset contains 2,018 orders from April to September 2026: 1,444 delivered, 573 RTO and 1 in-transit. The RTO rate among completed orders is approximately 28.4%.

Recommendation: use a risk-based approach. Confirm uncertain orders, encourage prepaid payment for selected orders, and avoid automatic cancellation unless there is strong evidence of repeated RTO behaviour.

2. Key Dataset Findings

• Overall RTO: 573 out of 2,017 completed orders were returned, an RTO rate of approximately 28.4%.

• First-time customers: Preliminary analysis showed an RTO rate of approximately 30.3%. First-time customers should not automatically be treated as risky.

• Repeated RTO history: Customers with two previous RTOs showed an RTO rate of approximately 76.5% in a small group of 34 orders. This is a warning signal, but the small sample size requires monitoring.

The preliminary city-level analysis showed higher RTO rates in Gorakhpur (63%) and Darbhanga (58.1%). Treat these as monitoring indicators, not reasons to reject every order from those locations.

3. Recommended Decision Rules

The engine uses four actions: Ship as-is for orders without major risk signals; Ship after WhatsApp confirmation for first-time customers or uncertain details; Nudge to prepaid for selected higher-value orders with a 5% discount; and Hold/cancel for customers with two or more previous RTOs, subject to monitoring.

Thresholds are editable in the application configuration. Name and address checks are basic rule-based checks; they do not prove that a customer or address is fake.

4. Financial Trade-off

The estimated cost of one RTO is INR 169.40: forward shipping INR 65, return shipping INR 65, packaging and handling INR 25, and expected damage cost INR 14.40 (8% × INR 180).

The business must balance this cost against delivered gross margin and customer acquisition cost. The stated CAC is INR 300 or more, so cancelling genuine first-time customers can also create a business loss.

Prioritise confirmation and prepaid nudges before cancellation. This protects genuine customers while helping reduce avoidable RTO costs.

5. September 2026 Retrospective Simulation

After removing duplicate order IDs, the September 2026 dataset contains 329 unique orders.

Decision	Orders
Ship as-is	44
Ship after confirmation	262
Nudge to prepaid	20
Hold/cancel	3

All three orders in the hold/cancel category had historically ended in RTO. At INR 169.40 per order, potential RTO cost avoidance is INR 508.20. No historically delivered orders were placed in this category, so estimated genuine-order gross margin loss is INR 0. Estimated net impact: positive INR 508.20.

These are retrospective estimates based on historical outcomes, not proof of actual savings or guaranteed future results.

6. Risks and Monitoring

Some genuine customers may have incomplete addresses or unusual names. To reduce false positives, use WhatsApp confirmation before cancellation where possible; monitor false positives and customer responses; review city-level patterns rather than blocking locations; and track prepaid conversion and discount impact.

The WhatsApp flow is simulated and is not connected to a real WhatsApp account. The prototype uses rule-based checks rather than a live LLM service.

7. Next Two Weeks

Week 1: Test the engine on historical orders, review flagged orders and verify English and Telugu confirmation messages.

Week 2: Run a small monitored pilot, compare confirmation responses, prepaid conversions and RTO outcomes, and adjust thresholds based on results.

Final Recommendation

Do not automatically cancel all risky COD orders. Start with WhatsApp confirmation and prepaid nudges, and reserve hold/cancel for stronger risk signals. This offers a better balance between reducing RTO costs and protecting genuine customers.